Bilkent University

Department of Electrical and Electronics Engineering

EEE-313 Project Third Part Report

Thermocouple Instrumentation Amplifier Controlled Heater

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[ID]

"I affirm that I have not given or received any unauthorized help on this report and that this work is my own."

Introduction:

The aim of this project is to create a single-supply thermocouple amplifier that will generate a voltage difference proportional to the temperature at the output. Using a K-type thermocouple, OPAMPS, a BJT and a LED to show the situation. The thermocouple will create small voltage differences with the temperature. The difference will be amplified by an OPAMP circuit. Then, the circuit will make the heater switch from ON and OFF. This way, the temperature of the resistor will be kept stable. The resistor will be kept at 36℃ since $T_s = 30 + mod([ID]\ ,40)$℃ where [ID] is my Bilkent ID. When the heater is ON, the LED must turn ON for visual feedback.

Here are the specifications for this project:

1) **The output voltage should be 2V∓0.5V when the input voltage is 0V (the termocouple at room temperature).**

2) **The output voltage should be 9V∓1V when the input voltage is at $39.2\mu Vx36 = 1.4112mV$ (temperature is at T_S).**

3) **LED should turn OFF when voltage passes 9V and it should be ON otherwise.**

We had a design phase. In this part, I designed my circuit in LTSpice and analyzed my circuit. After that, I drew my schematic on Diptrace. When I made sure my schematic is correct, I corverted into a Diptrace PCB Layout. There, I connected and placed my components in the right places where all the components fit on a 2100milx2100mil PCB layer. In the figures below, we can see these parts.

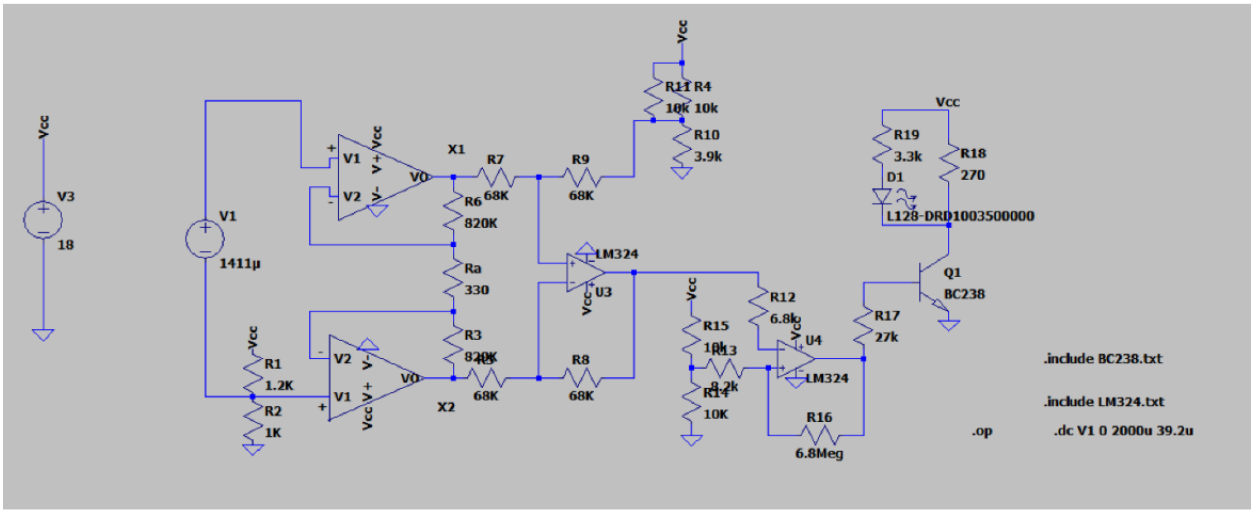
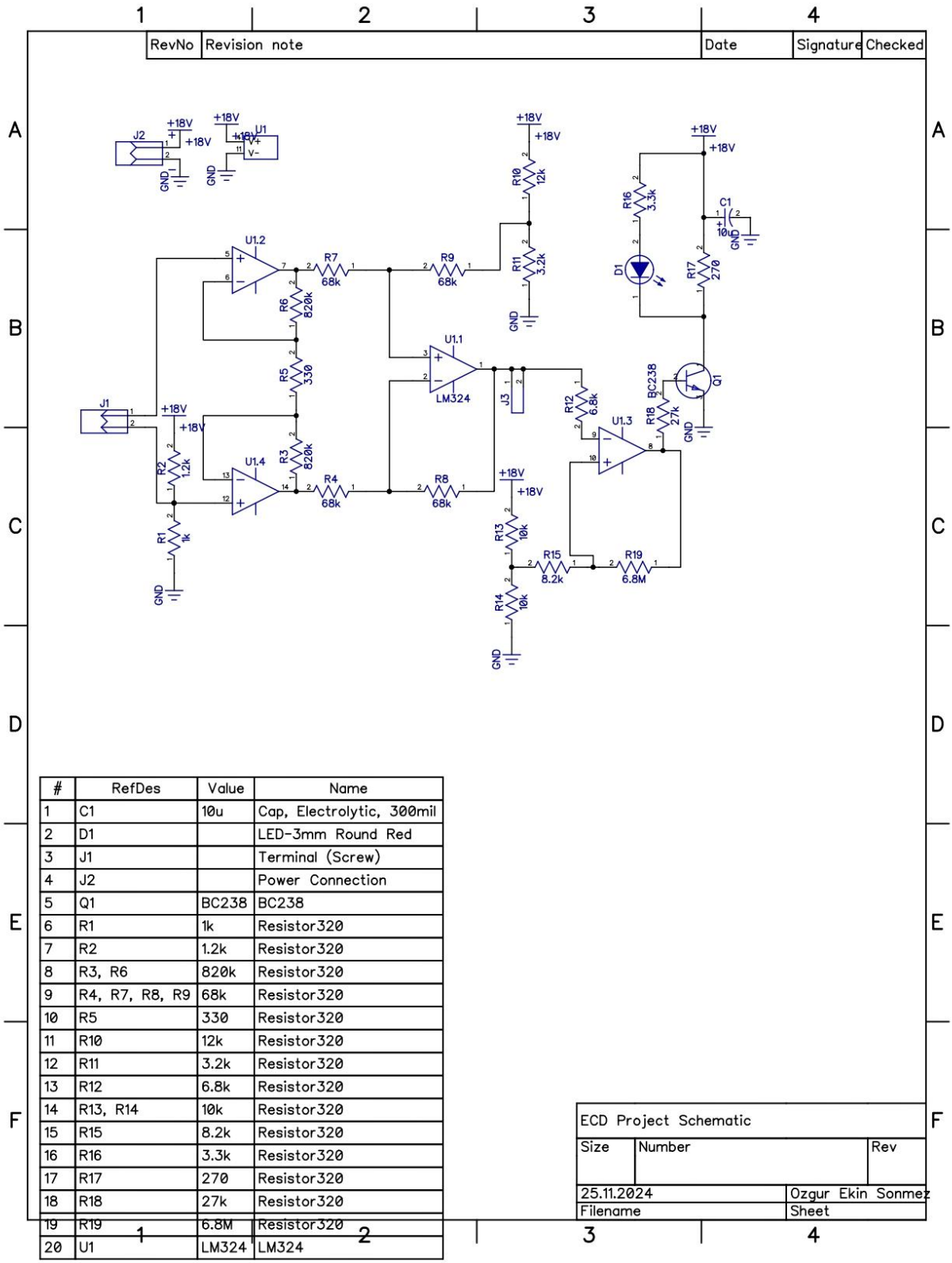


Figure 1: LTSpice Design of the Circuit



#	RefDes	Value	Name
1	C1	10u	Cap, Electrolytic, 300mil
2	D1		LED-3mm Round Red
3	J1		Terminal (Screw)
4	J2		Power Connection
5	Q1	BC238	BC238
6	R1	1k	Resistor320
7	R2	1.2k	Resistor320
8	R3, R6	820k	Resistor320
9	R4, R7, R8, R9	68k	Resistor320
10	R5	330	Resistor320
11	R10	12k	Resistor320
12	R11	3.2k	Resistor320
13	R12	6.8k	Resistor320
14	R13, R14	10k	Resistor320
15	R15	8.2k	Resistor320
16	R16	3.3k	Resistor320
17	R17	270	Resistor320
18	R18	27k	Resistor320
19	R19	6.8M	Resistor320
20	U1	LM324	LM324

ECD Project Schematic		
Size	Number	Rev
25.11.2024		Ozgur Ekin Sonmez
Filename		Sheet

Figure 2: Diptrace Schematic

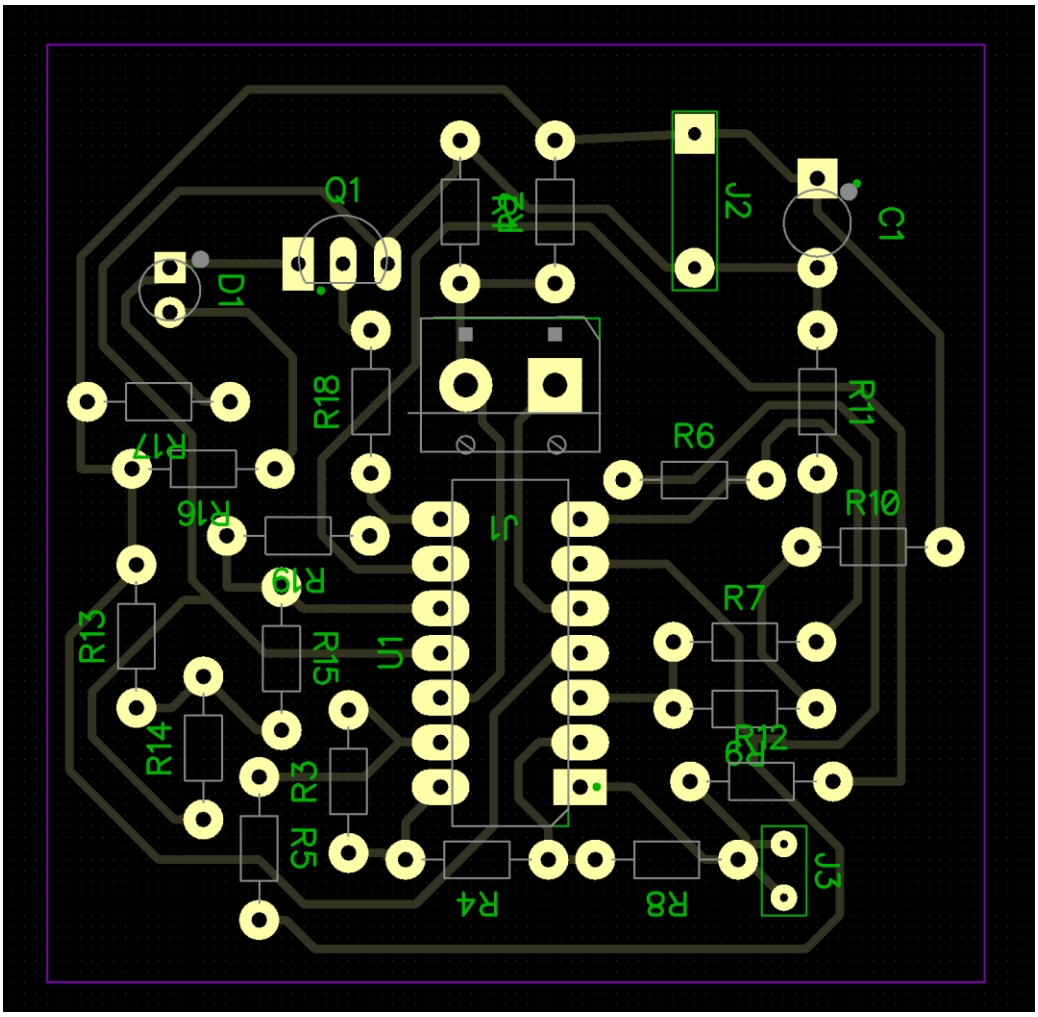


Figure 3: Diptrace PCB Layout

After these, I created the Gerber files which are top silk, bottom and board files. By using PCB Way OnlineGerberView we can inspect out PCB. Here are those images below:

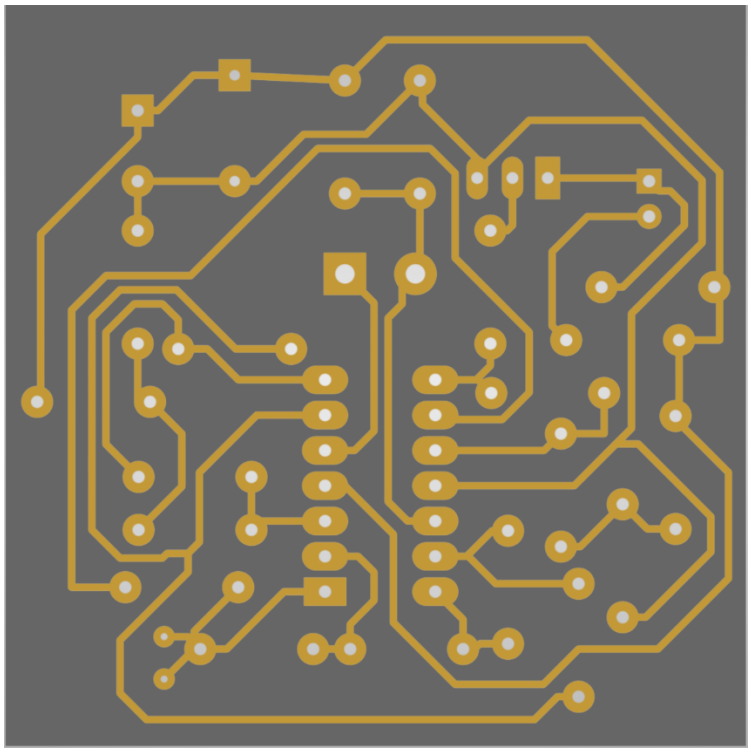


Figure 4: PCB Bottom View

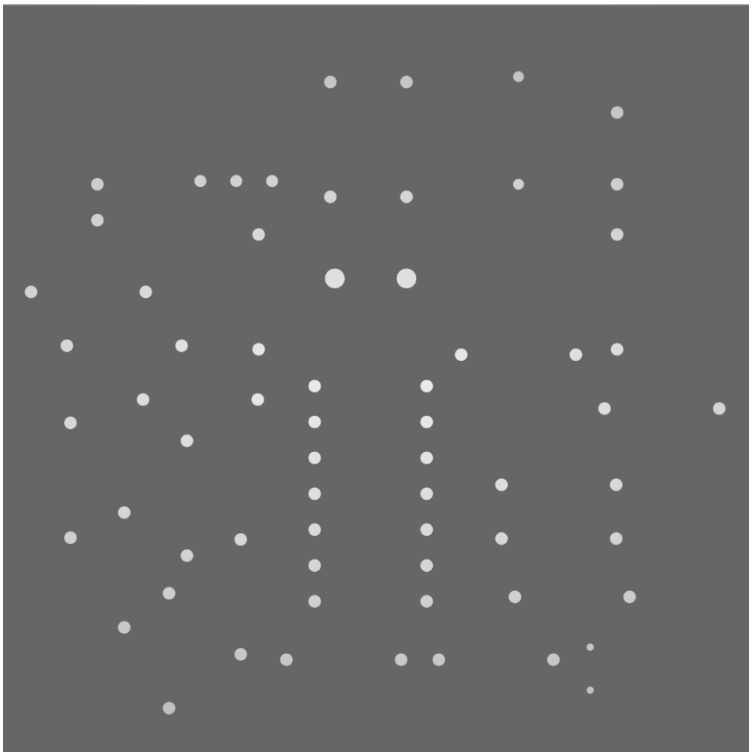


Figure 5: PCB Top View

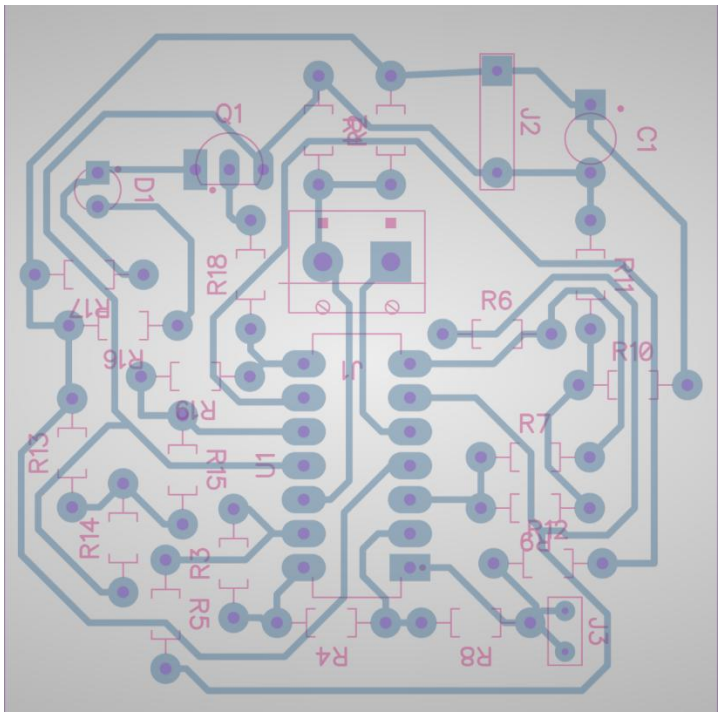


Figure 6: Layers of PCB

After these it is time to show the experimental results.

Experimental Results:

To begin, I soldered my components on my PCB. Before that, I tried my circuit on a breadboard to make sure it is working. When I made sure, I started to solder each component one by one. Here is my PCB after the soldering.

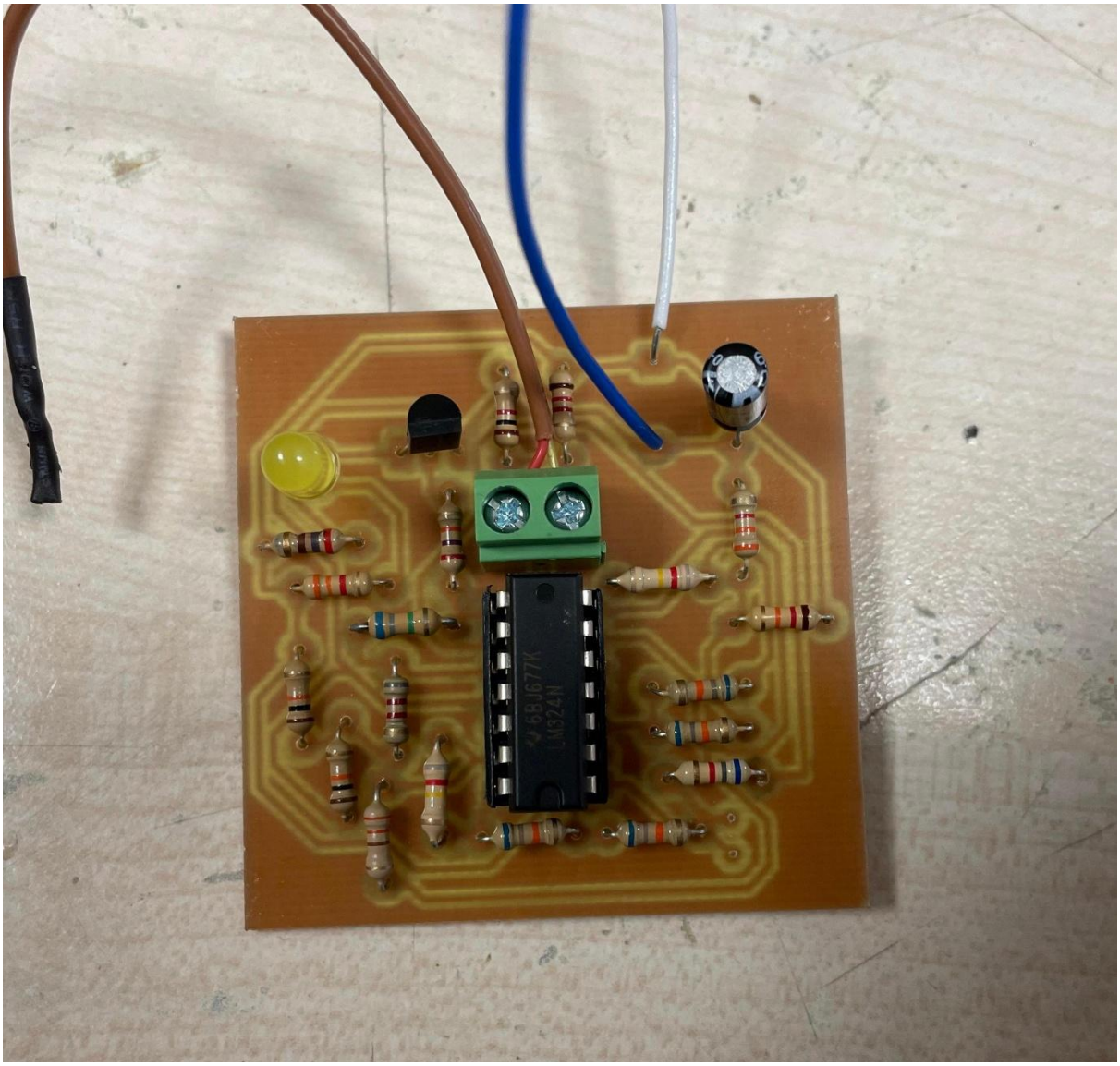


Figure 7: Experimental Circuit with components soldered.

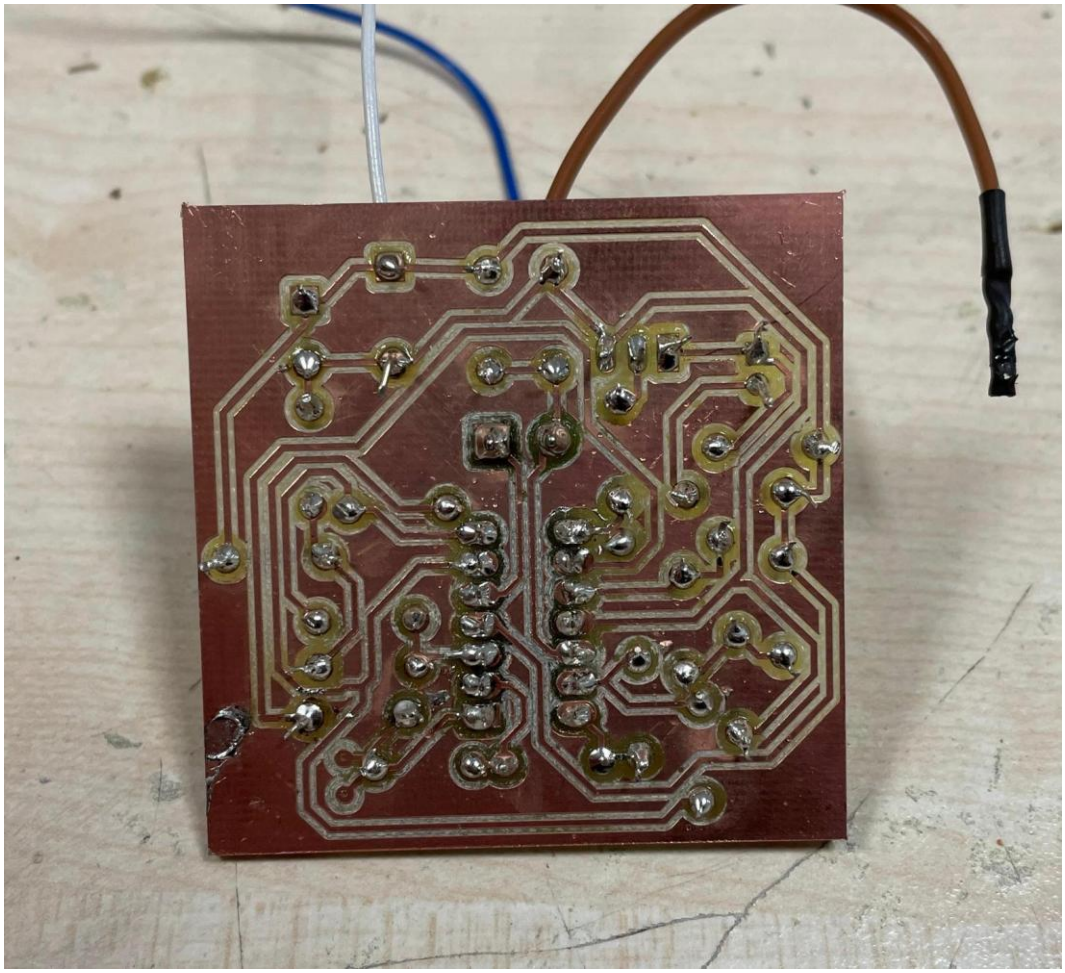


Figure 8: Experimental Circuit From the Back

1) **The output voltage should be 2V∓0.5V when the input voltage is 0V (the termocouple at room temperature).**

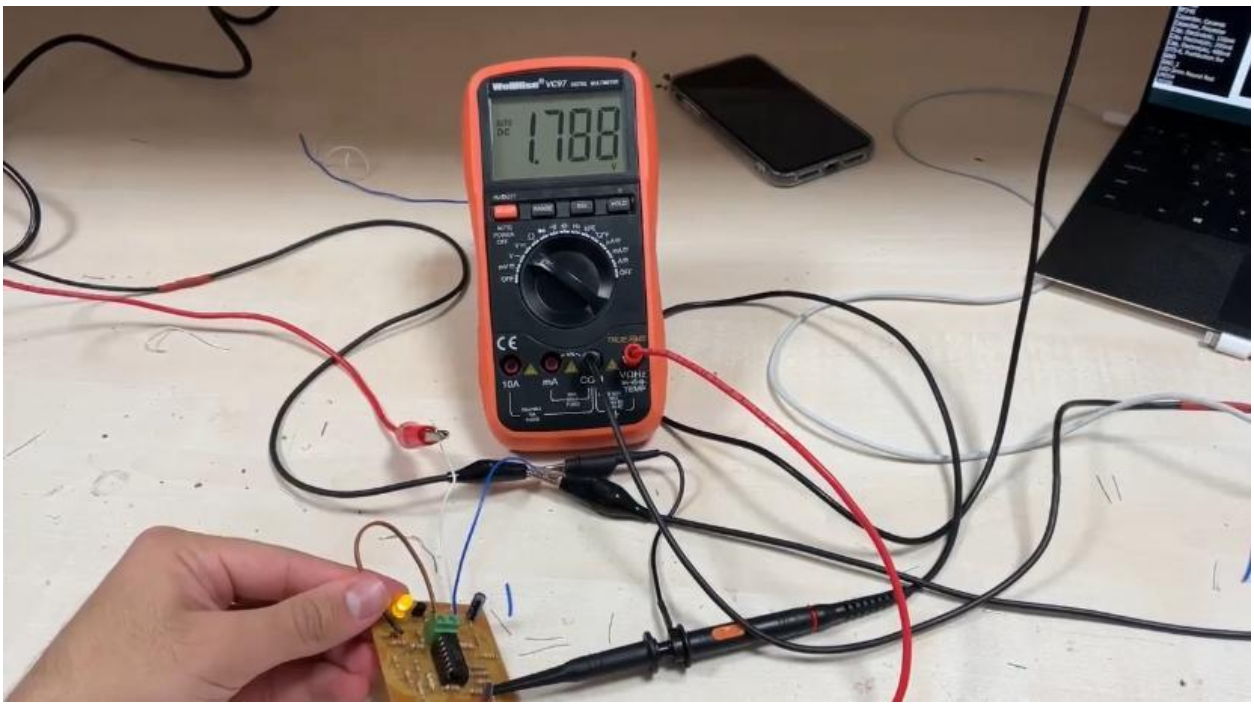


Figure 9: Vout=1.788V at room temperature.

As it can be seen from the figure, the output voltage at room temperature is 1.788V which is around the limit of this specification. Therefore, this value satisfies the given specification.

2) **The output voltage should be 9V∓1V when the input voltage is at $39.2\mu V x 36 = 1.4112mV$ (temperature is at T_S).**

3) **LED should turn OFF when voltage passes 9V and it should be ON otherwise.**

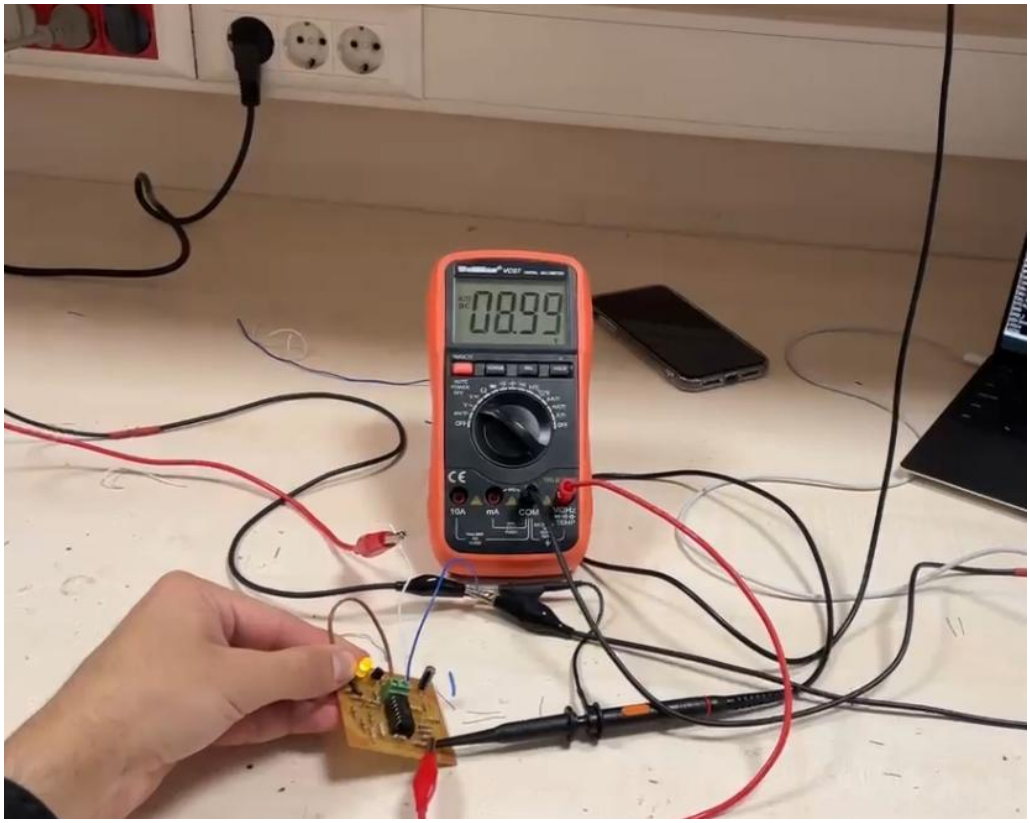


Figure 10: When Vout=8.99V LED is ON

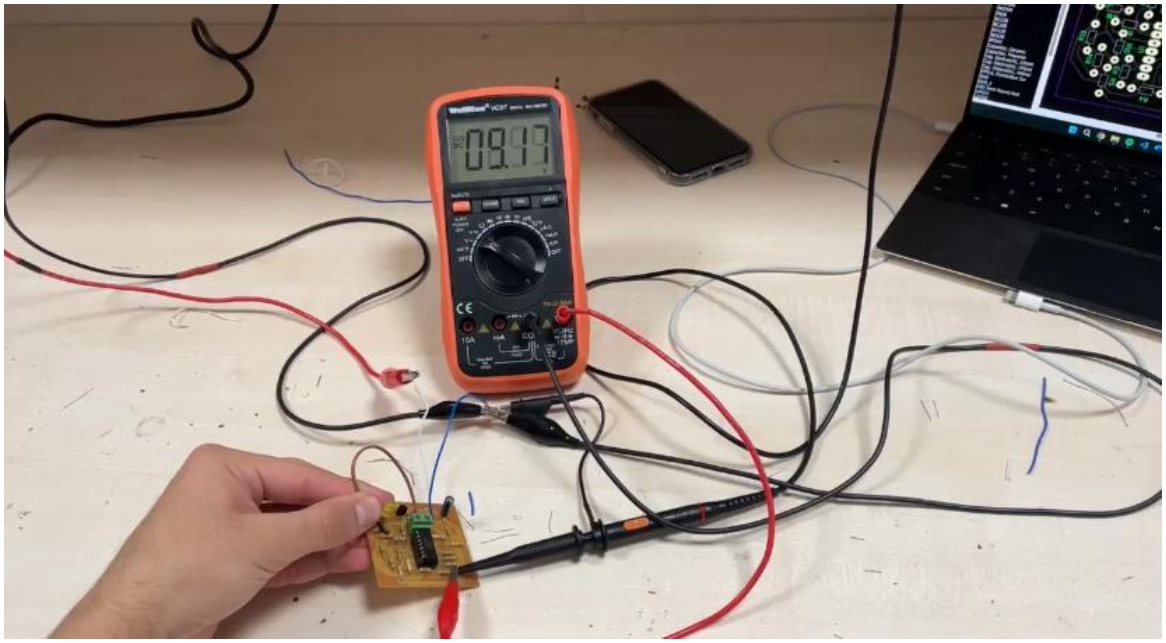


Figure 11: When Vout=9.17V LED is OFF

From these 2 figures we can clearly see that it satisfies the second and third specifications. On Figure 10, we are measuring Vout=8.99V and the LED is ON. When it passes 9V, on Figure 11, we are seeing that when Vout=9.17V the LED turns OFF and current through the load resistor stopped passing through it. This will result in a decrease of temperature, therefore the output voltage will try to go back to 9V, and will eventually get stabilized around 9V. When it is being heated (Vout is smaller than 9V), the LED is always ON. For example, the heater will be OFF when it reaches to 9V. The multimeter measures a little bit higher than 9V at some of the first switch OFFs because of the physics, it cannot cool down really fast. When it is cooling down, the LED is OFF and the output voltage will start to decrease. These are enough show that both second and third requirements were satisfied. Here is an extra image to show the PCB close up when the LED is ON.

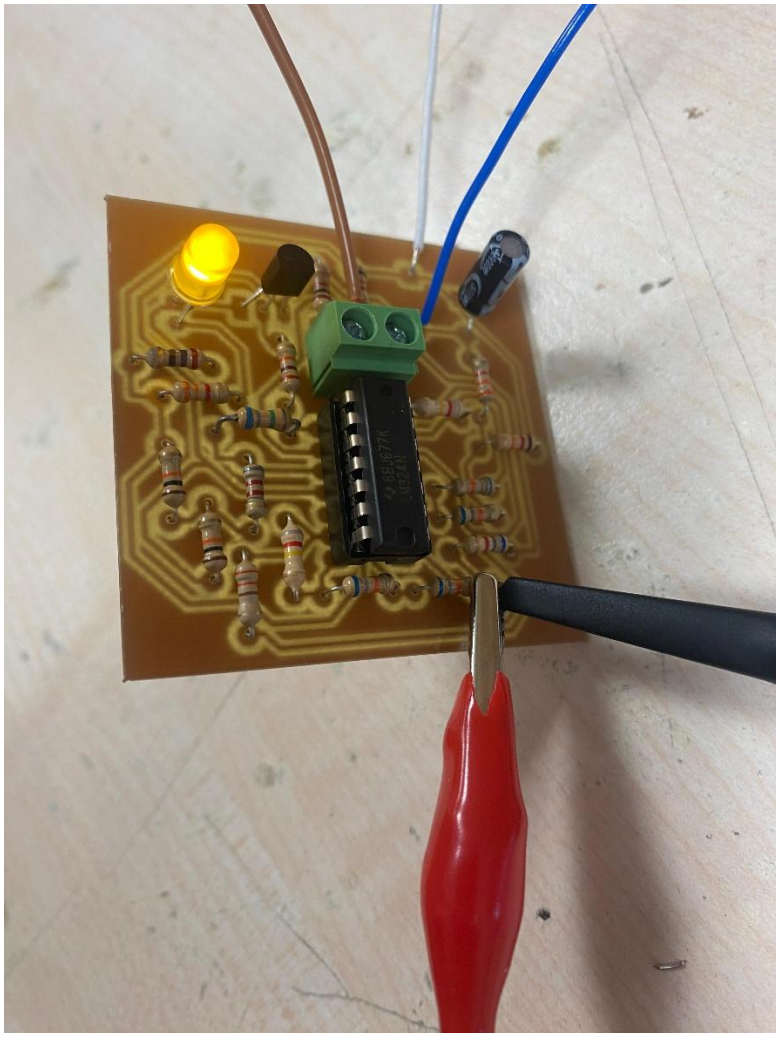


Figure 12: LED is ON

Lastly, I checked the oscillations on the output when the resistor is being heated. Here is an image of the oscilloscope.

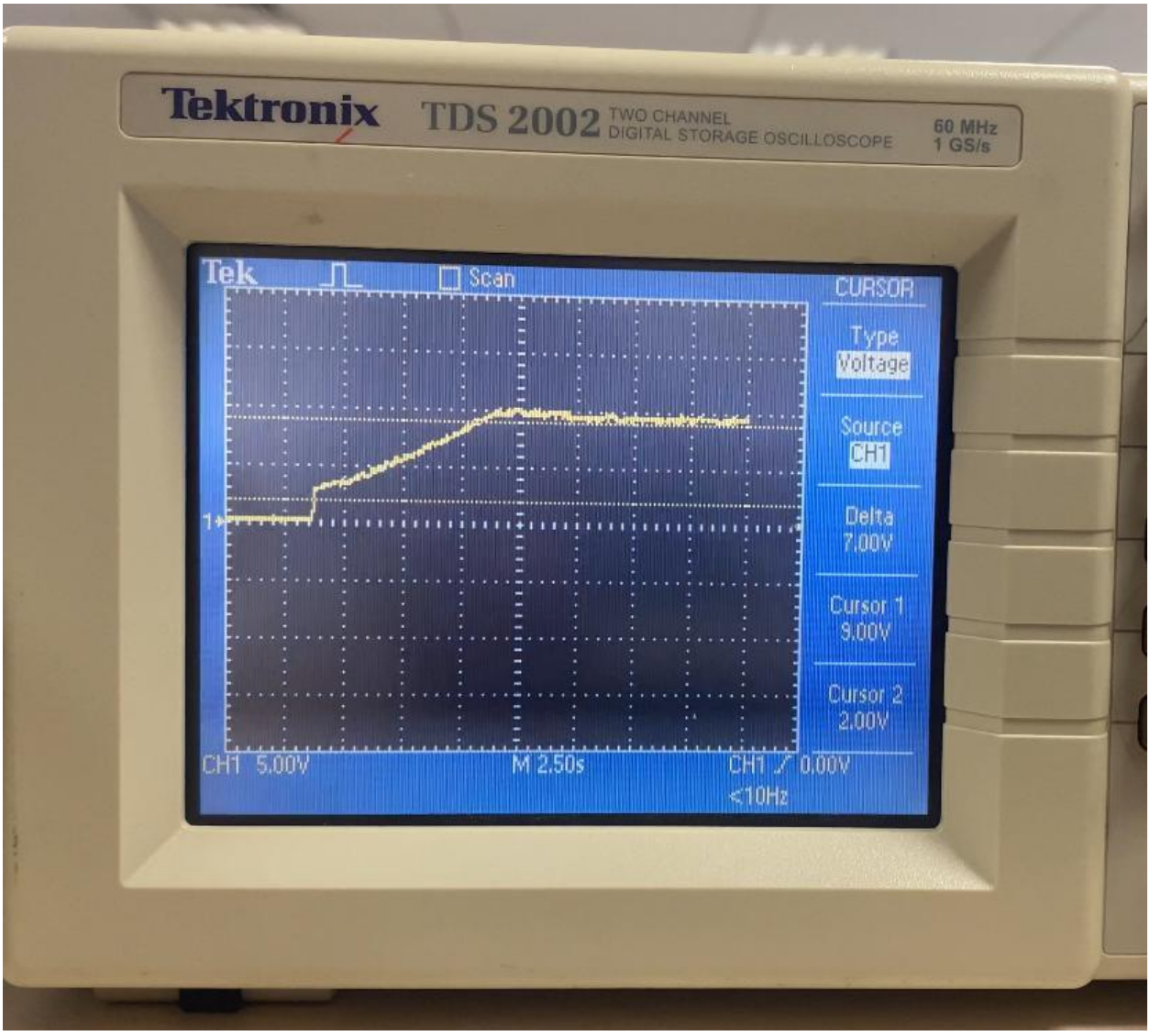


Figure 13: Vout from the oscilloscope.

As it can be seen from Figure 13, there are 2 cursors on 2V and 9V. When the power supply is on, we can see an instant increase from 0 to close to 2V. After that we can see a increase where the heater is ON. After it reaches 9V, heater closes itself and will go ON and OFF where the output value is getting closer and closer to 9V.

Conclusion

To conclude, out aim was to design a Thermocouple Instrumentation Amplifier Controlled Heater and try to implement it on a PCB which we designed. Through this Project, I learned how to design a PCB in most optimal way for my purposes, improved my skills on the components that I used. General concept of K-type thermocouples, using BJTs as switches, and instrumentation amplifier circuits were learned. Overall, this Project was successfull with small errors. These errors may be caused by human error, the small error on the standard components, or maybe even the solder itself.

Youtube Video:

https://www.youtube.com/watch?v=gqMhvIIfWEo